Quiz 1

Name: ID:

1) Software development is difficult primarily because the problem is usually:

A) Well-defined

B) Ambiguous

C) Simple

D) Non-existent

2) The requirements in software development are characterized as:

A) Static and clear

B) Unclear and changing

C) Clear from the beginning

D) Irrelevant

3) The domain in which the software problem exists is referred to as:

A) Solution domain

B) Application domain

C) Development domain

D) Problem domain

4) What is a crucial first step in problem-solving within software development?

A) Writing code immediately to address the problem.

B) Understanding the problem thoroughly.

C) Choosing the programming language to use.

D) Documenting the potential solutions.

5) the scientific method does not provide clear guidance on:

A) How to conduct experiments

B) Where to pick up ideas and hypotheses

C) How to write research papers

D) How to analyze data

6) What is the definition of software Engineering?

A collection of techniques, methodologies, and tools that help with the production of a high quality software system developed with a given budget before a given deadline while changes occur.

7) Define a system model in the context of software engineering

A system model is an integrated representation that combines the object model, the functional model and the dynamic model to provide a comprehensive description of the software system

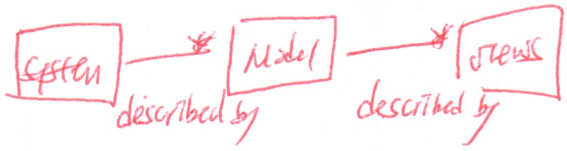
8) What are the two special types of hierarchies in a class diagram?

1) Kind of

2) part of

Quiz 1

9) Please draw the relationships between systems, models, and views in a class diagram

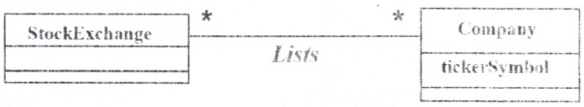


10) The Java code provided below is meant to be a direct translation of the accompanying class diagram. What should be written inside the rectangle in the class diagram to ensure it matches the Java code?

Java Code

```
public class StockExchange {
    public m_Company = new Vector();
};
public class Company {
  public int m_tickerSymbol;
  public Vector m_StockExchange = new Vector();
};
```

Class Diagram



11) Sequence diagrams are useful for illustrating:

A) The workflow and the steps involved in a process

B) The functional behavior of the system as seen by the user

C) The dynamic behavior between objects of the system

D) The static relationships between classes

12) Activity diagrams are best suited for describing:

A) The static structure of the system

B) The functional behavior of the system from the user's perspective

C) The dynamic behavior of a system, particularly the workflow

D) The attributes and associations of objects within the system

13) What should the focus be on in a textual use case description?

flow of event

14) UML sequence diagrams are useful for:

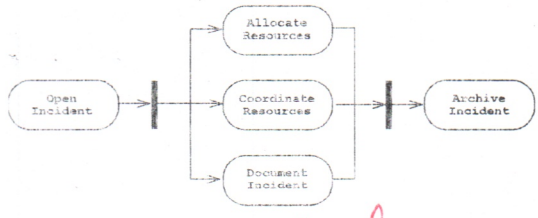
A) Speeding up the development process

B) Identifying or finding missing objects

C) Representing physical components

D) Generating code automatically

15) Describe the role of the black synchronization bar (join) and the black splitting bar (fork)



Represents points of concurrency management. It allows for parallel execution of activities.